

# KYLE BINDER

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## APPOINTMENTS

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| <b>Louisiana State University</b><br><i>Postdoctoral Researcher</i> | 2025 – Present |
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## EDUCATION

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| <b>The Ohio State University</b><br><i>M.S. &amp; Ph.D. in Mathematics</i> | 2019 – 2025<br>Advisor: Eric Katz |
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| <b>Baylor University</b><br><i>B.A. in Mathematics and Linguistics (University Scholars)</i> | 2015 – 2019 |
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## RESEARCH INTERESTS

Combinatorial algebraic geometry, combinatorics, toric varieties, matroids, vector bundles, tropical geometry.

## PAPERS AND PREPRINTS

1. *The Top-Nonvanishing Weight Cohomology of a Smooth Toric Variety*, in preparation.  
(~15 pp., expected October 2026)
2. *Augmented Singular Cohomology, Uniform Matroids, and Real-Rootedness* (with Lorenzo Vecchi), in preparation. (~20 pp., expected September 2026)
3. *The Singular Cohomology Ring of a Uniform Matroid: Combinatorics and Lefschetz Properties*, submitted, [arXiv:2606.07490](https://arxiv.org/abs/2606.07490). 21 pp.
4. *The Singular Cohomology Ring of a Matroid*, submitted, [arXiv:2412.05732](https://arxiv.org/abs/2412.05732). 35 pp.
5. *The Unipotent Tropical Fundamental Group* (with Eric Katz), to appear in J. Lond. Math. Soc., [arXiv:2406.13973](https://arxiv.org/abs/2406.13973). 55 pp.
6. *Limit Sets and Internal Transitivity in Free Group Actions* (with Jonathan Meddaugh), preprint, [arXiv:1908.07382](https://arxiv.org/abs/1908.07382). 17 pp.
7. *On 2-coloring the integers to forbid a distance and monochromatic translates of a triangle*, *Geombinatorics* **27** (2018), no. 4, 137–146.

## TALKS

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| <i>The Singular Cohomology Ring of a Matroid</i><br>Brown Combinatorics Seminar | November 2026 |
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| <i>The Singular Cohomology Ring of a Matroid</i><br><a href="#">Combinatorics &amp; Geometry BLT Seminar</a> | September 2026 |
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| <i>A Tropical Approach to the Unipotent Fundamental Group of a Hyperplane Arrangement Complement</i><br>Hyperplane Arrangements and Polytopes Special Session<br>AMS Southeastern Sectional Meeting | October 2025 |
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| <i>Cohomology of Toric Varieties from Matroids</i><br>LSU Geometry and Topology Seminar | September 2025 |
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| <i>The Tropical Unipotent Fundamental Group</i><br>5 Minute Talk at Tropical Geometry: Moduli Spaces and Matroids Workshop               | October 2024   |
| <i>Higher Tor Groups of Matroids</i><br>5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA)                          | May 2024       |
| <i>Higher Tor Groups of Matroids</i><br>OSU Combinatorics and Probability Seminar                                                        | April 2024     |
| <i>Higher Tor Groups of Stanley–Reisner Rings of Matroids</i><br>5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA) | May 2023       |
| <i>The Tropical Unipotent Fundamental Group</i><br>20 Minute Talk at Workshop on Matroids and Tropical Combinatorics (MaTroCom)          | January 2023   |
| <i>A Survey of Tropical Cohomology</i><br>OSU Algebraic Geometry Seminar                                                                 | September 2022 |
| <i>The Geometries of Hyperplane Arrangement Complements</i><br>OSU Student Algebraic Geometry Seminar                                    | April 2022     |
| <i>p-adic Integration on Hyperelliptic Curves</i><br>OSU Student Algebraic Geometry Seminar                                              | February 2021  |
| POSTERS                                                                                                                                  |                |
| <i>The Singular Cohomology Ring of a Matroid</i><br>Combinatorics at the Confluence                                                      | July 2026      |
| <i>The Singular Cohomology Ring of a Matroid</i><br>Texas Algebraic Geometry Symposium                                                   | April 2026     |
| <i>The Singular Cohomology Ring of a Matroid</i><br>Arizona–New Mexico Symposium on Commutative Algebra                                  | December 2025  |
| <i>What is the Cohomology of a Non-Compact Toric Variety?</i><br>Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures                    | December 2023  |
| TEACHING                                                                                                                                 |                |
| LSU Math 4997 – Vertically Integrated Research: Combinatorial Topology<br><i>Instructor</i>                                              | Fall 2026      |
| LSU Math 1550 – Calculus 1<br><i>Instructor</i>                                                                                          | Fall 2025      |
| OSU MATH 1131 – Business Calculus<br><i>Recitation Instructor</i>                                                                        | Spring 2024    |
| OSU MATH 1131 – Business Calculus<br><i>Recitation Instructor</i>                                                                        | Autumn 2022    |
| OSU MATH 1148 – College Algebra<br><i>Recitation Instructor</i>                                                                          | Autumn 2021    |
| OSU MATH 1148 – College Algebra<br><i>Recitation Instructor, Virtual (Zoom)</i>                                                          | Autumn 2020    |

## ORGANIZATION

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| Special Session on Algebraic and Geometric Aspects of Matroid Theory<br>at AMS Central Sectional Meeting<br><i>Co-organizer</i> | October 2026   |
| LSU Geometry and Topology Seminar<br><i>Co-organizer</i>                                                                        | 2025 – Present |
| OSU Directed Reading Program (DRP)<br><i>Co-organizer</i>                                                                       | 2021 – 2025    |
| Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures<br><i>Local Organizer</i>                                                  | December 2023  |